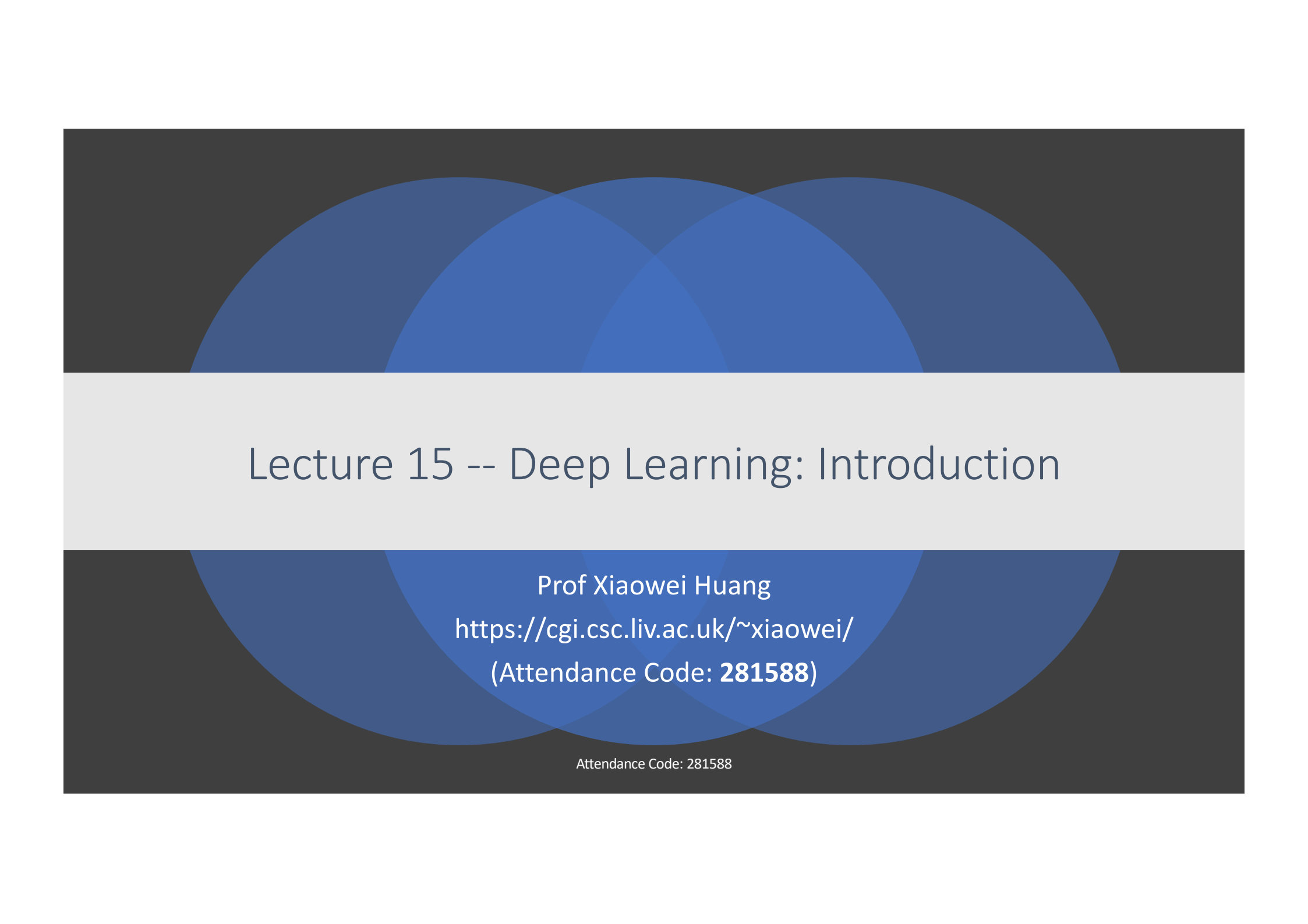




Lecture 15 -- Deep Learning: Introduction

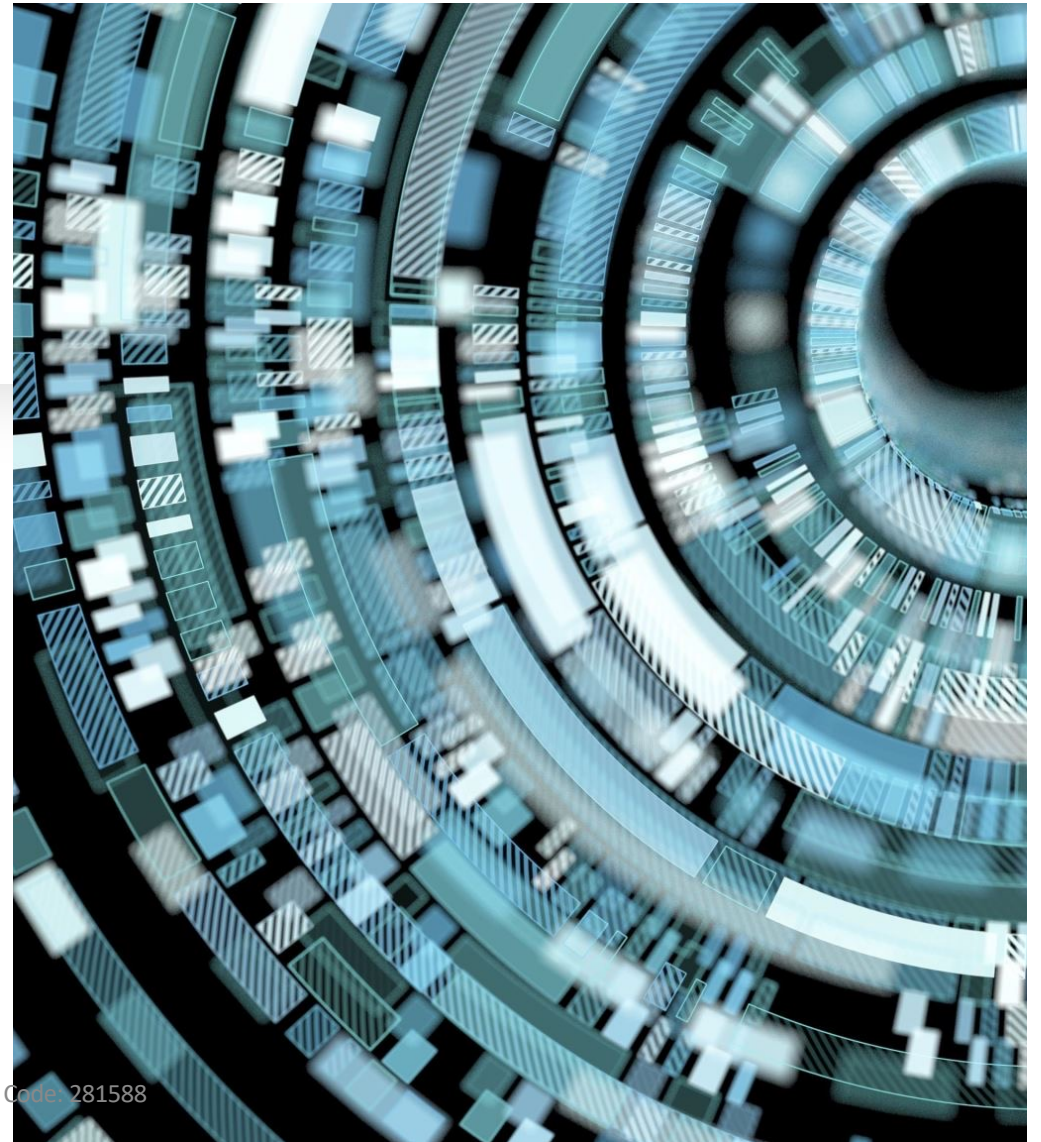
Prof Xiaowei Huang
<https://cgi.csc.liv.ac.uk/~xiaowei/>
(Attendance Code: **281588**)

Attendance Code: 281588

Up to now, we have included:

- Basic machine learning knowledge
- Four traditional machine learning algorithms
 - Decision tree.
 - K-nearest neighbor
 - Linear regression
 - Naïve bayes
- Adversarial attack and defence
- cf. [10 Algorithms: Machine Learning Engineers Need to Know](#)
- cf. [The 10 Best Machine Learning Algorithms for Data Science Beginners](#)

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What will be included in the fourth component?

Deep learning

- Introduction
- Functional view & features
- Forward and backward computation
- CNNs
- RNNs
- Transformer

Model evaluation



Today's topics

- Deep Learning in
 - Computer Vision
 - Natural Language Processing (NLP)
 - Speech
 - Robotics and AI
 - Music and the arts!
- A brief history of Neural Networks

Deep Learning in Computer Vision



Object and activity recognition

https://www.youtube.com/watch?v=qrzQ_AB1DZk



Object detection and segmentation

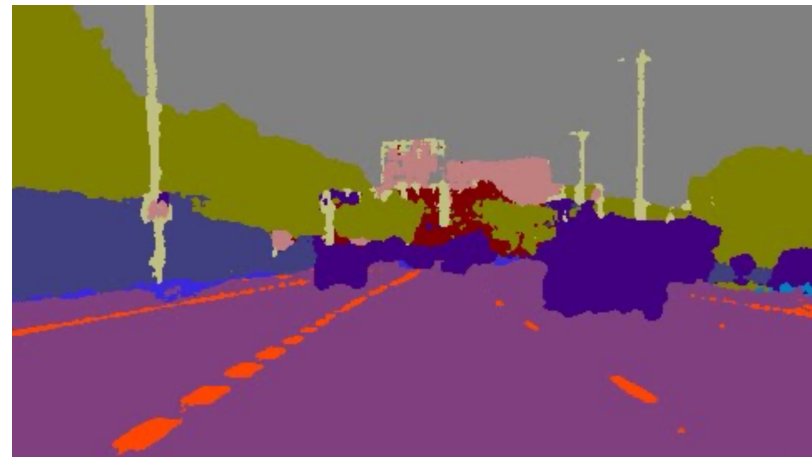
https://www.youtube.com/watch?v=CxanE_W46ts



Image captioning and Q&A

<https://www.youtube.com/watch?v=8BFzu9m52sc>

Deep Learning in Computer Vision



Why should we be impressed?

Vision is ultra challenging!

- For a small 256x256 resolution and for 256 pixel values
- a total $2^{524,288}$ of possible images
- In comparison there are about 10^{24} stars in the universe

Visual object variations

- Different viewpoints, scales, deformations, occlusions

Semantic object variations

- Intra-class variation
- Inter-class overlaps

Deep Learning in Robotics



Self-driving cars

<https://www.youtube.com/watch?v=-96BEoXJMs0>



Drones and robots

<https://www.youtube.com/watch?v=2hGngG64dNM>

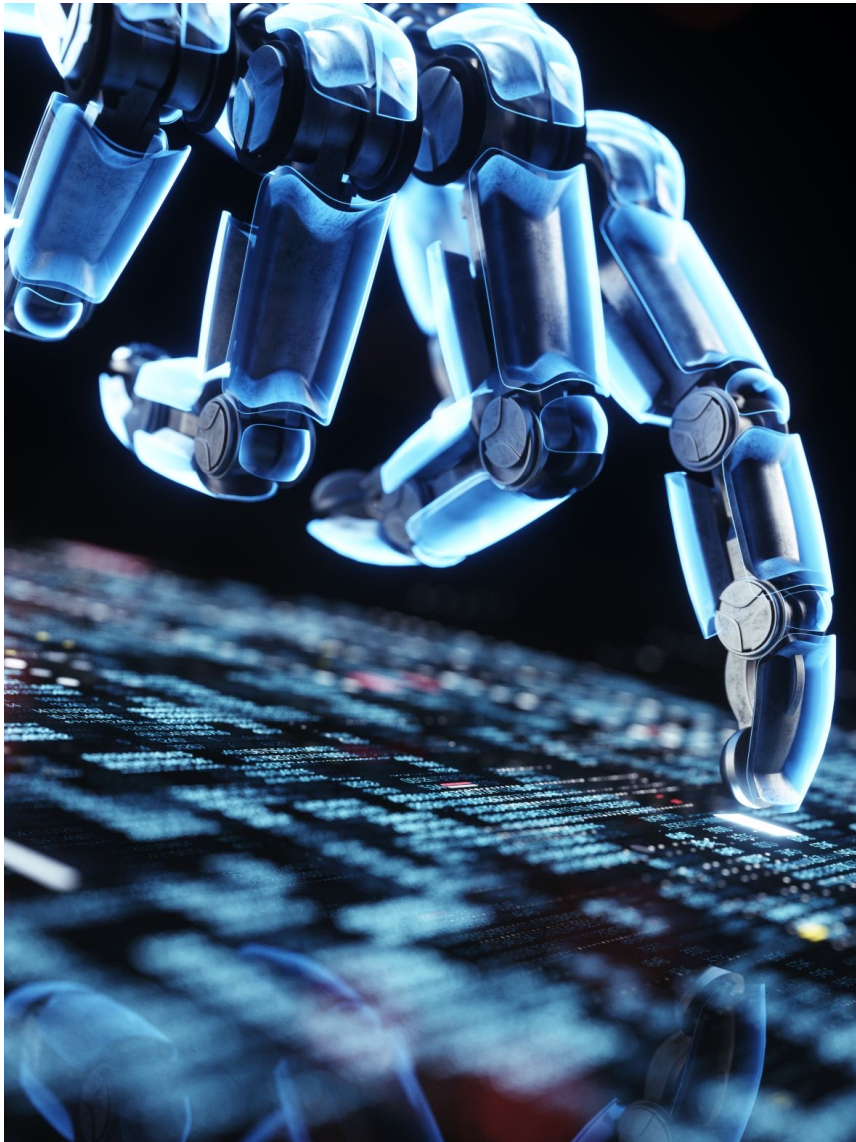


Game AI

<https://www.youtube.com/watch?v=V1eYniJ0Rnk>

Deep Learning in Robotics





Why should we be impressed?

- Typically, robotics are considered in controlled environments
 - Laboratory settings, Predictable positions, Standardized tasks (like in factory robots)
- What about real life situations?
 - Environments constantly change, new tasks need to be learnt without guidance, unexpected factors must be dealt with
- Game AI
 - At least $10^{10^{48}}$ possible GO games. Where do we even start?

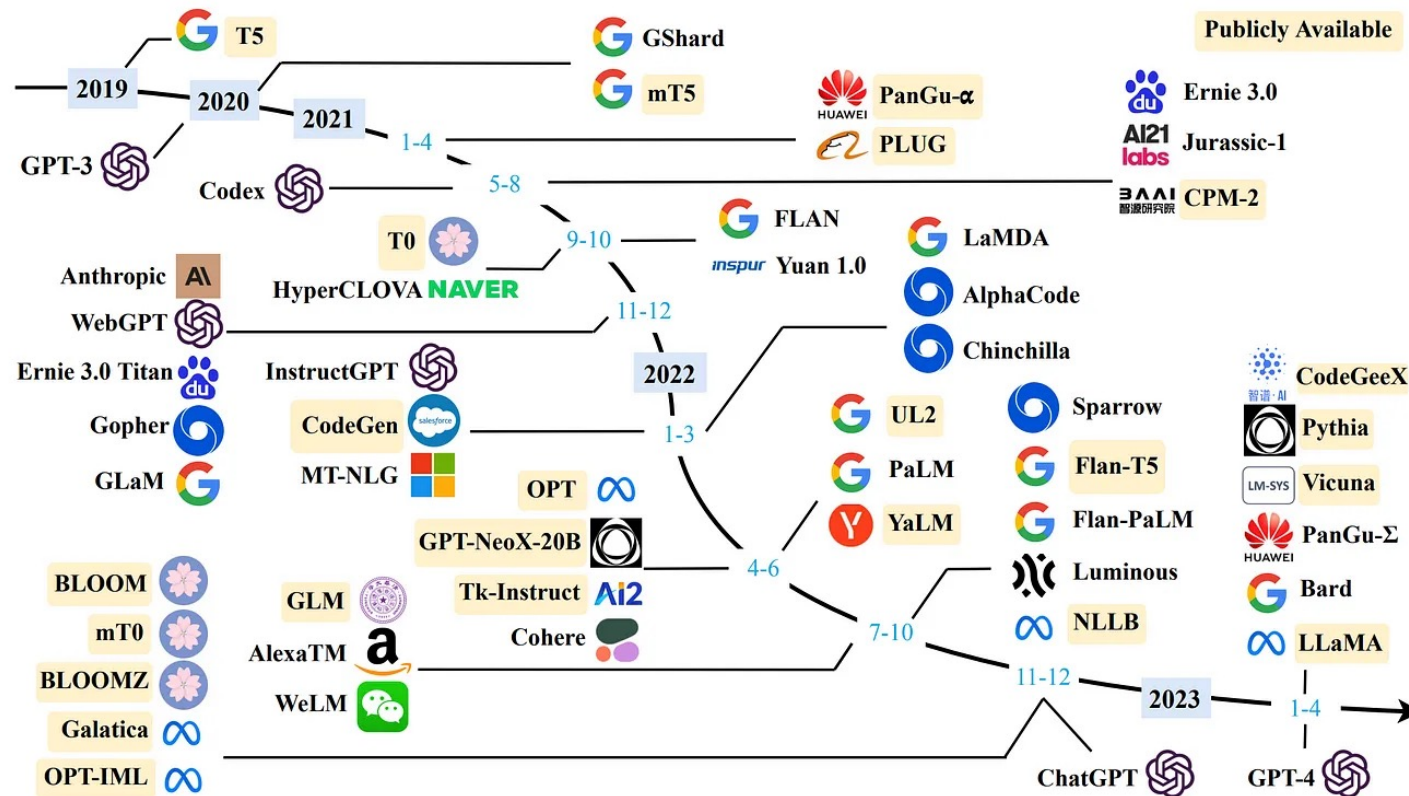
Deep Learning in NLP and Speech

- Word and sentence representations

Newspapers			
New York San Jose	New York Times San Jose Mercury News	Baltimore Cincinnati	Baltimore Sun Cincinnati Enquirer
NHL Teams			
Boston Phoenix	Boston Bruins Phoenix Coyotes	Montreal Nashville	Montreal Canadiens Nashville Predators
NBA Teams			
Detroit Oakland	Detroit Pistons Golden State Warriors	Toronto Memphis	Toronto Raptors Memphis Grizzlies
Airlines			
Austria Belgium	Austrian Airlines Brussels Airlines	Spain Greece	Spainair Aegean Airlines
Company executives			
Steve Ballmer Samuel J. Palmisano	Microsoft IBM	Larry Page Werner Vogels	Google Amazon

Table 2: Examples of the analogical reasoning task for phrases (the full test set has 3218 examples). The goal is to compute the fourth phrase using the first three. Our best model achieved an accuracy of 72% on this dataset.

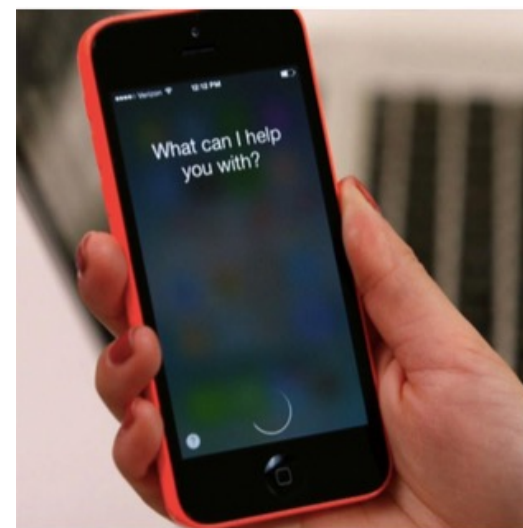
Large Language Models



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Deep Learning in NLP and Speech

- Speech recognition and Machine translation



And more recently, GPT-4o:

https://youtu.be/vgYi3Wr7v_g?si=qYJ6wHvC5C0ns_RO

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Why should we be impressed?

NLP is an extremely complex task

- synonymy (“chair”, “stool” or “beautiful”, “handsome”)
- ambiguity (“I made her duck”, “Cut to the chase”)

NLP is very high dimensional

- assuming 150K english words, we need to learn 150K classifiers
- with quite sparse data for most of them

Beating NLP feels the closest to achieving true AI

- although true AI probably goes beyond NLP, Vision, Robotics, ... alone

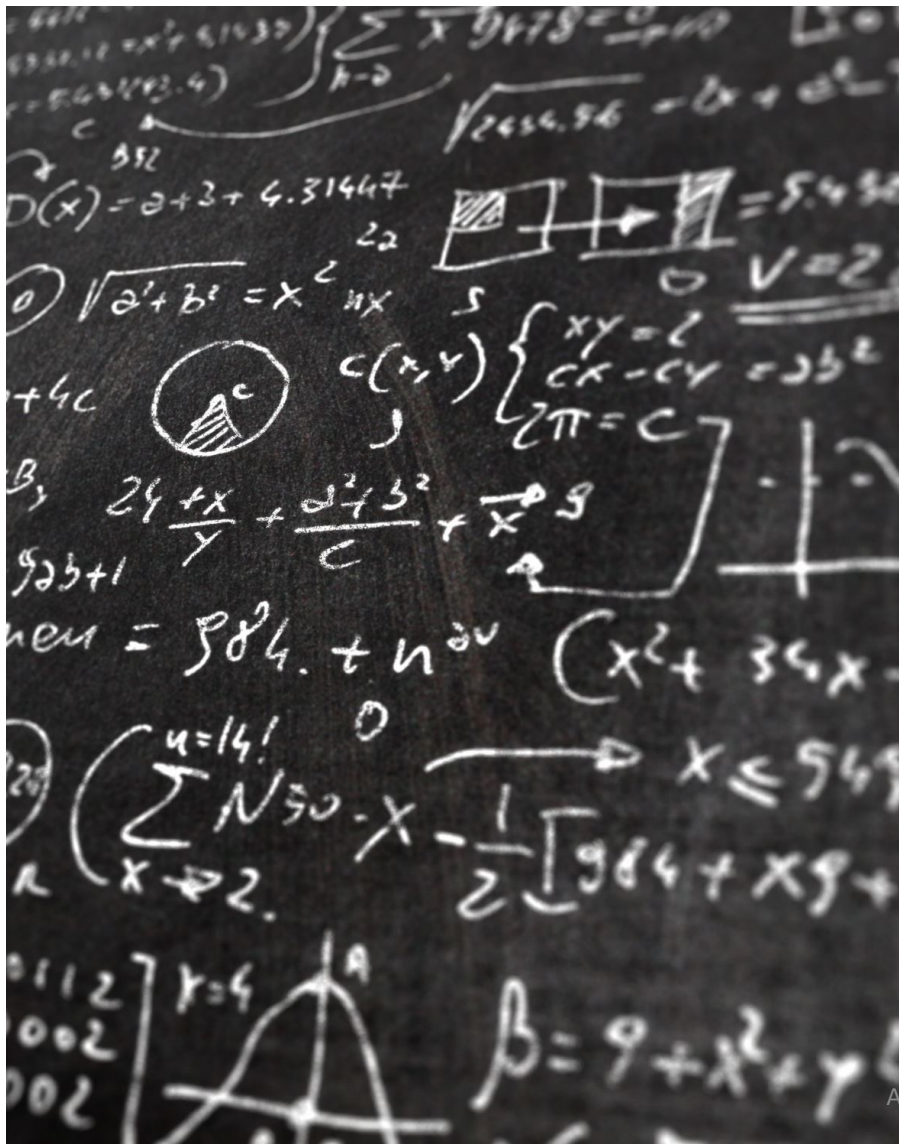
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Deep Learning in the arts

- Imitating famous painters



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Handwriting

- <http://www.cs.toronto.edu/~graves/handwriting.html>

Text --- up to 100 characters, lower case letters work best

How can I make you fail?

Style --- either let the network choose a writing style at random or prime it with a real sequence to make it mimic that writer's style.

- ☐ Take the words away when they are
- ☐ He dismissed the idea
- ☐ prison welfare Officer complement
- ☐ She looked closely as she
- ☐ at Hinkcombe is being adopted for
- ☒ random style

Bias --- increasing the bias makes the samples more legible but less diverse. Using a high bias *and* a priming sequence makes the network write in a neater version of the original style.

Samples

3

Scale

Linewidth

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Why should we be impressed?

Music, painting, etc. are tasks that are uniquely human

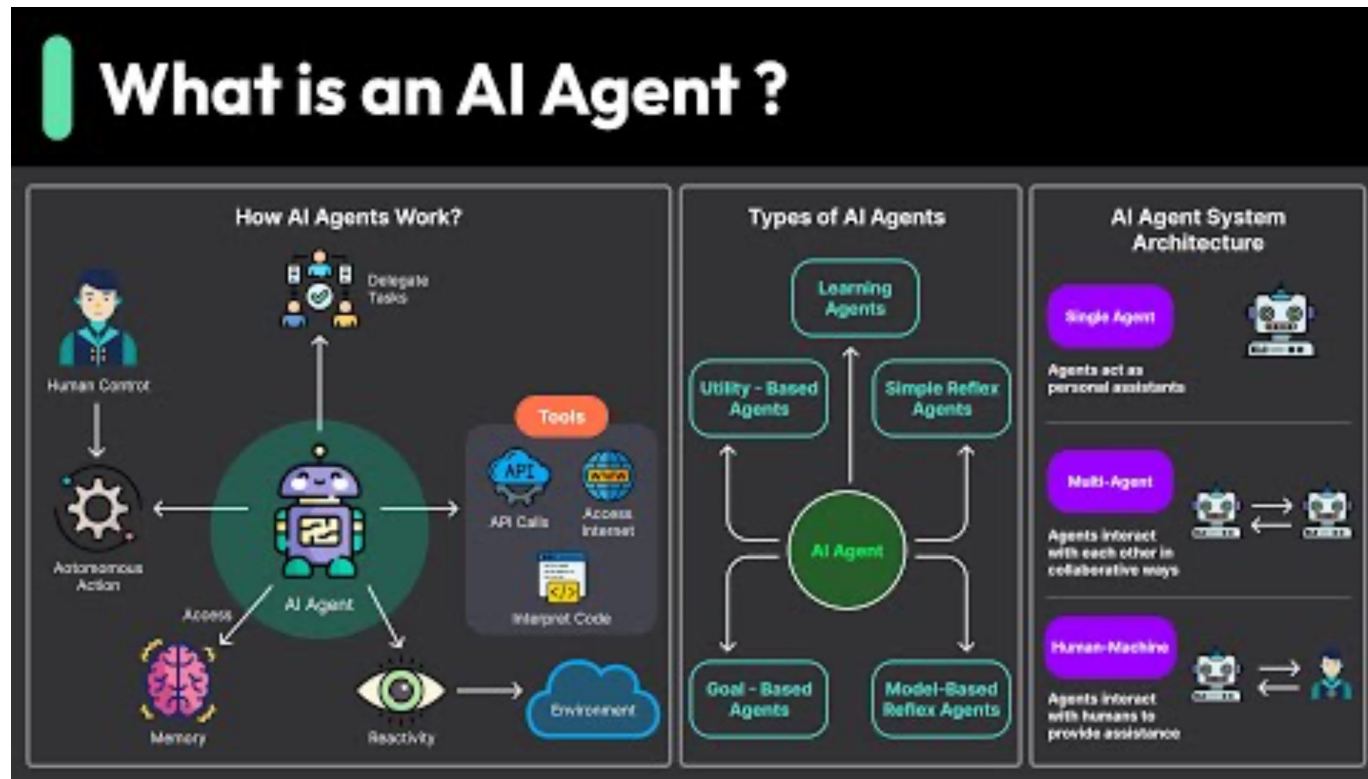
- Difficult to model
- Even more difficult to evaluate (if not impossible)

If machines can generate novel pieces even remotely resembling art, they must have understood something about “beauty”, “harmony”, etc.

Have they really learned to generate new art, however?

- Or do they just fool us with their tricks?

AI Agent



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A brief history of Neural Networks & Deep Learning

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First Appearance (Roughly)

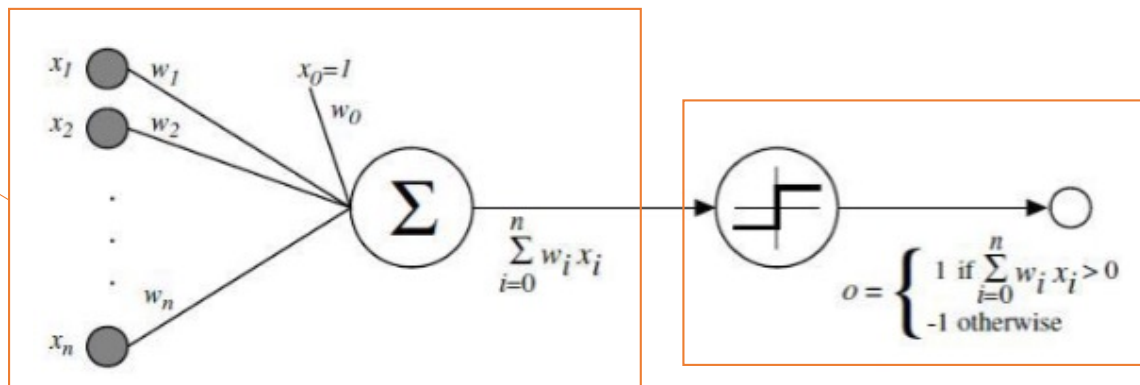


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Perceptrons

- Rosenblatt proposed a machine for binary classifications
- Main idea
 - One weight w_i per input x_i
 - Multiply weights with respective inputs and add bias $x_0 = +1$
 - If result larger than threshold return 1, otherwise 0

Does this part
look similar as
 $w^T x + b$?



Linear classification?

Training a perceptron

- Rosenblatt's innovation was mainly the learning algorithm for perceptrons

- Learning algorithm

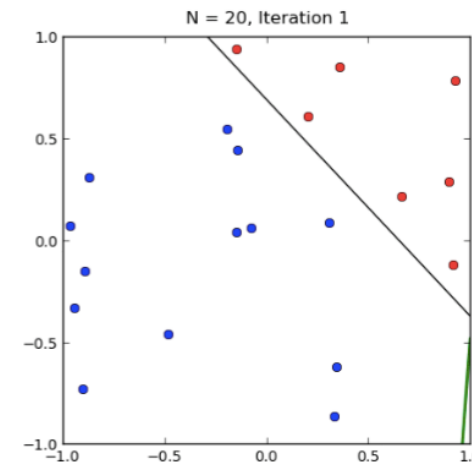
- Initialize weights randomly
- Take one sample x_i and predict y_i
- For erroneous predictions update weights
 - If the output was $\hat{y}_i = 0$ and $y_i = 1$, increase weights
 - If the output was $\hat{y}_i = 1$ and $y_i = 0$, decrease weights

$$\Delta w_i = \eta(y - \hat{y})x_i$$

$$w_i \leftarrow w_i + \Delta w_i$$

Repeat until no errors are made

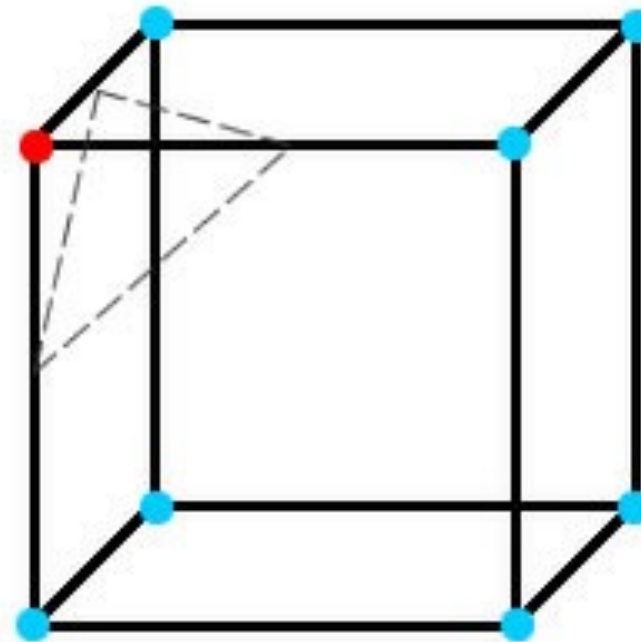
η is learning rate;
set to value $\ll 1$





Representational power of perceptrons

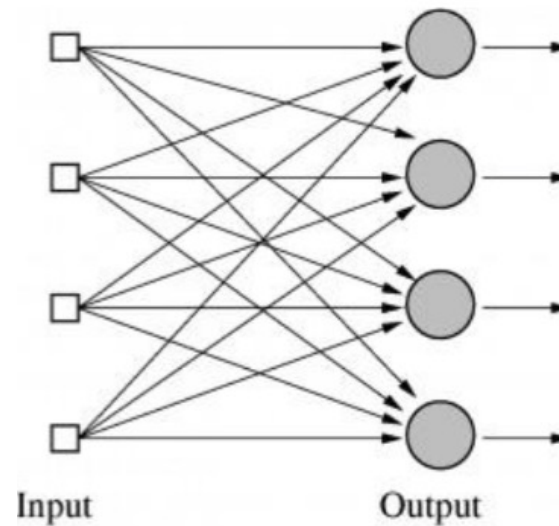
- ML models are approximating behaviors of complex functions.
- **Representation power** is related to ability of a ML model to assign proper labels to a particular instance and create well defined accurate decision boundaries for that class.
- In previous example, feature space was 2D so decision boundary was a line
- In higher dimensions, decision boundary is a hyperplane





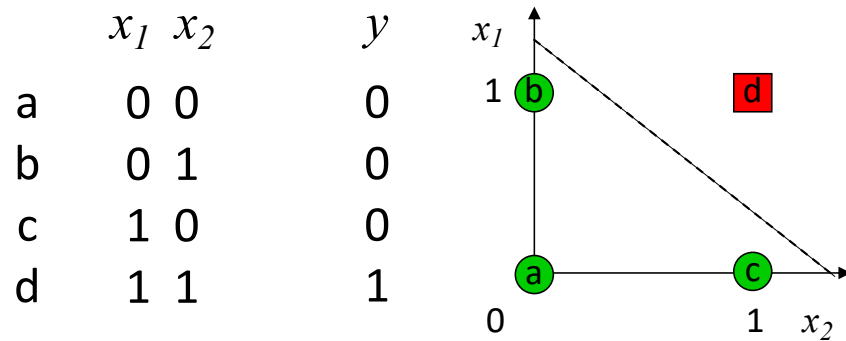
From a perceptron to a neural network

- One perceptron = one decision
- What about **multiple decisions**?
 - E.g. digit classification
 - Stack as many outputs as the possible outcomes into a layer
- Use one layer as input to the next layer
 - **Multi-layer** perceptron (MLP)
- Neural network

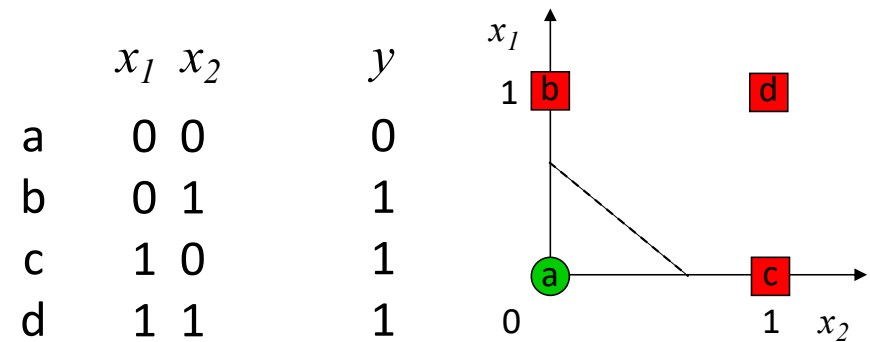


Some linearly separable functions

AND

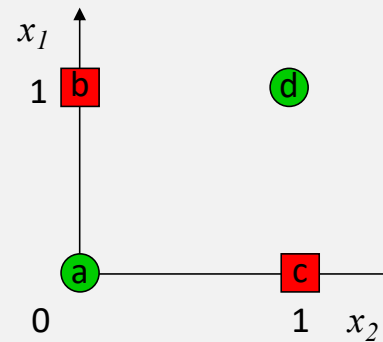


OR



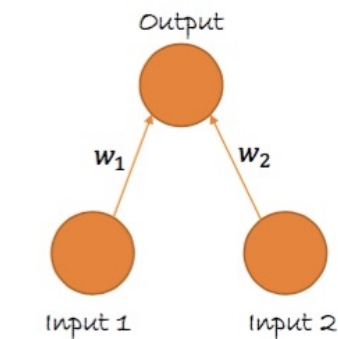
XOR is not linearly separable

	x_1	x_2	y
a	0	0	0
b	0	1	1
c	1	0	1
d	1	1	0



XOR & Multi-layer Perceptrons

- However, the exclusive or (XOR) cannot be solved by perceptrons
 - [Minsky and Papert, "Perceptrons", 1969]



Input 1	Input 2	Output
1	1	0
1	0	1
0	1	1
0	0	0

Assuming $w_1x_1 + w_2x_2 - \theta$

$0 w_1 + 0 w_2 < \theta \rightarrow 0 < \theta$

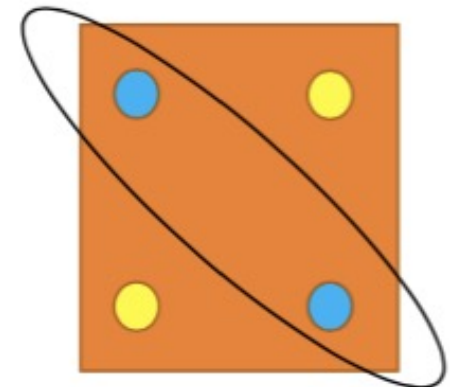
$0 w_1 + 1 w_2 > \theta \rightarrow w_2 > \theta$

$1 w_1 + 0 w_2 > \theta \rightarrow w_1 > \theta$

$1 w_1 + 1 w_2 < \theta \rightarrow w_1 + w_2 < \theta$

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inconsistent!!

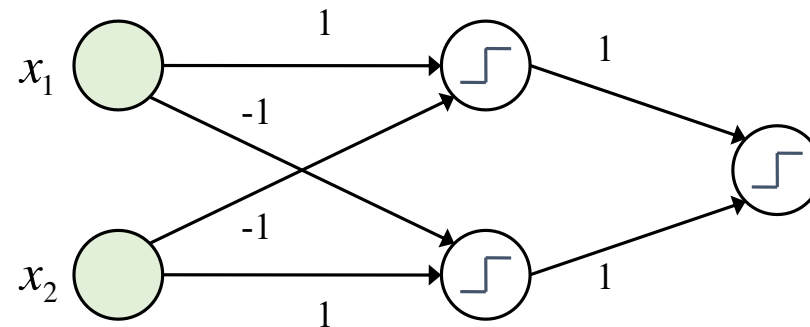


Minsky & Multi-layer perceptrons

- Interestingly, Minsky never said XOR cannot be solved by neural networks
 - Only that XOR cannot be solved with 1-layer perceptrons
- Multi-layer perceptrons can solve XOR
 - 9 years earlier Minsky built such a multi-layer perceptron

Input 1	Input 2	Output
1	1	0
1	0	1
0	1	1
0	0	0

A multilayer perceptron
can represent XOR



assume $w_0 = 0$ for all nodes

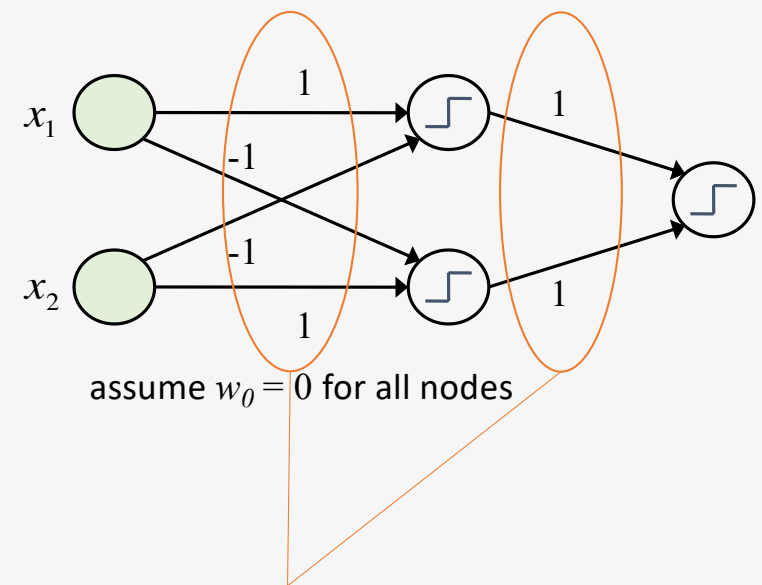
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Test to see if this is the case.

Minsky & Multi-layer perceptrons

- Interestingly, Minsky never said XOR cannot be solved by neural networks
 - Only that XOR cannot be solved with 1 layer perceptrons
- Multi-layer perceptrons can solve XOR
 - 9 years earlier Minsky built such a multi-layer perceptron
- However, how to train a multi-layer perceptron?
- Rosenblatt's algorithm not applicable, as it expects to know the desired target
 - For hidden layers we cannot know the desired target

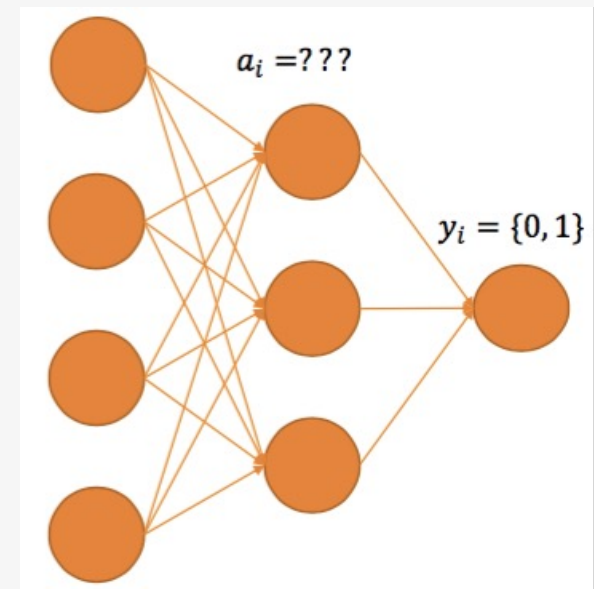
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How to obtain these parameters?

Minsky & Multi-layer perceptrons

- Interestingly, Minsky never said XOR cannot be solved by neural networks
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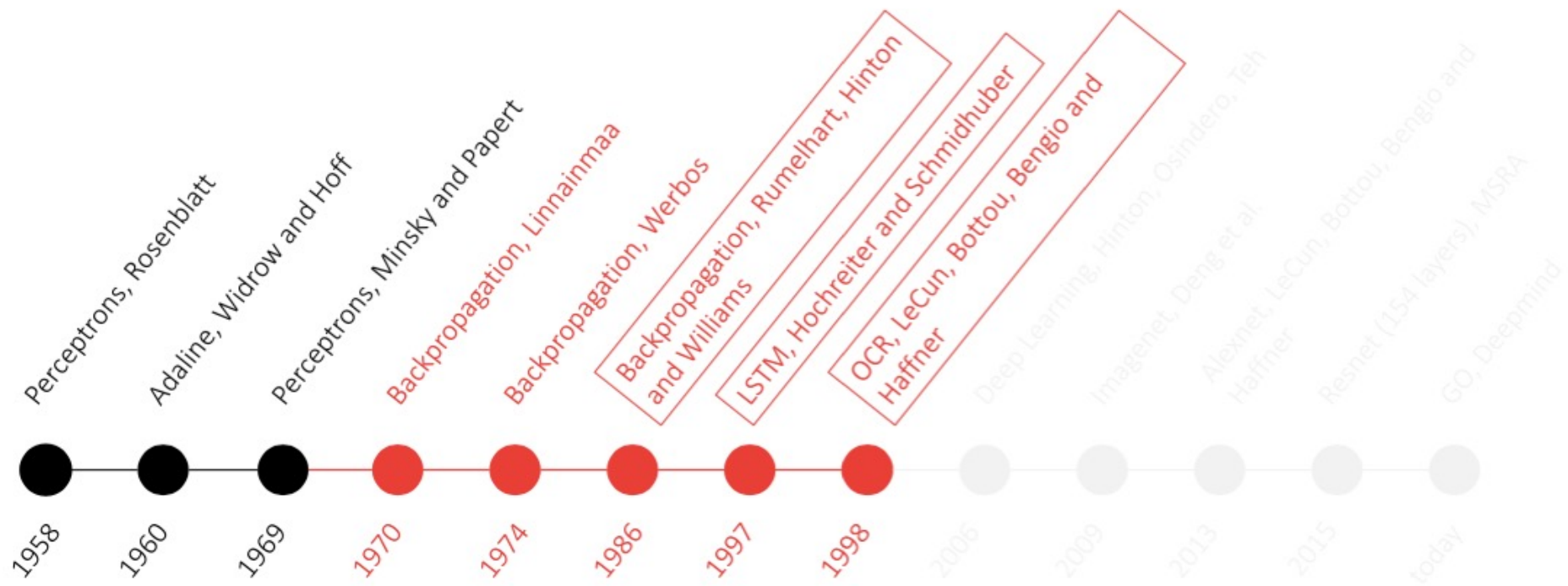




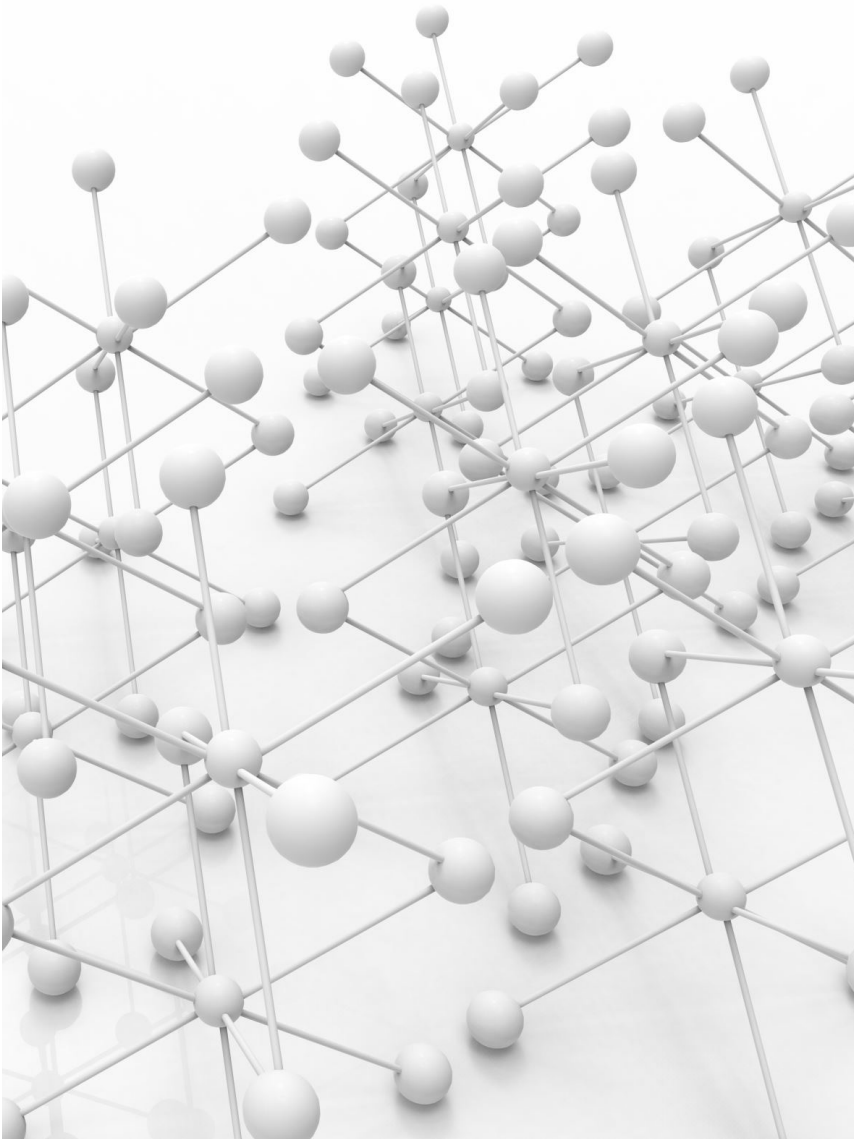
The first “AI winter”

- What everybody thought: “If a perceptron cannot even solve XOR, why bother? “
 - Also, the exaggeration did not help (walking, talking robots were promised in the 60s)
- As results were never delivered, further funding was slashed, neural networks were damned and AI in general got discredited
- “The AI winter is coming”
- Still, a few people persisted
- Significant discoveries were made, that laid down the road for today’s achievements

The “AI winter” despite notable successes



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Backpropagation

- Learning multi-layer perceptrons now possible
 - XOR and more complicated functions can be solved
- Efficient algorithm
 - Process hundreds of example without a sweat
 - Allowed for complicated neural network architectures
- Backpropagation still is the backbone of neural network training today
- Digit recognition in cheques (OCR) solved before the 2000



Recurrent networks

- Traditional networks are “too plain”
 - Static Input => Processing => Static Output
- What about dynamic input
 - Temporal data, Language, Sequences
- Memory is needed to “remember” state changes
 - Recurrent feedback connections
- What kind of memory
 - Long, Short?
 - Both! Long-short term memory networks (LSTM), Schmidhuber 1997

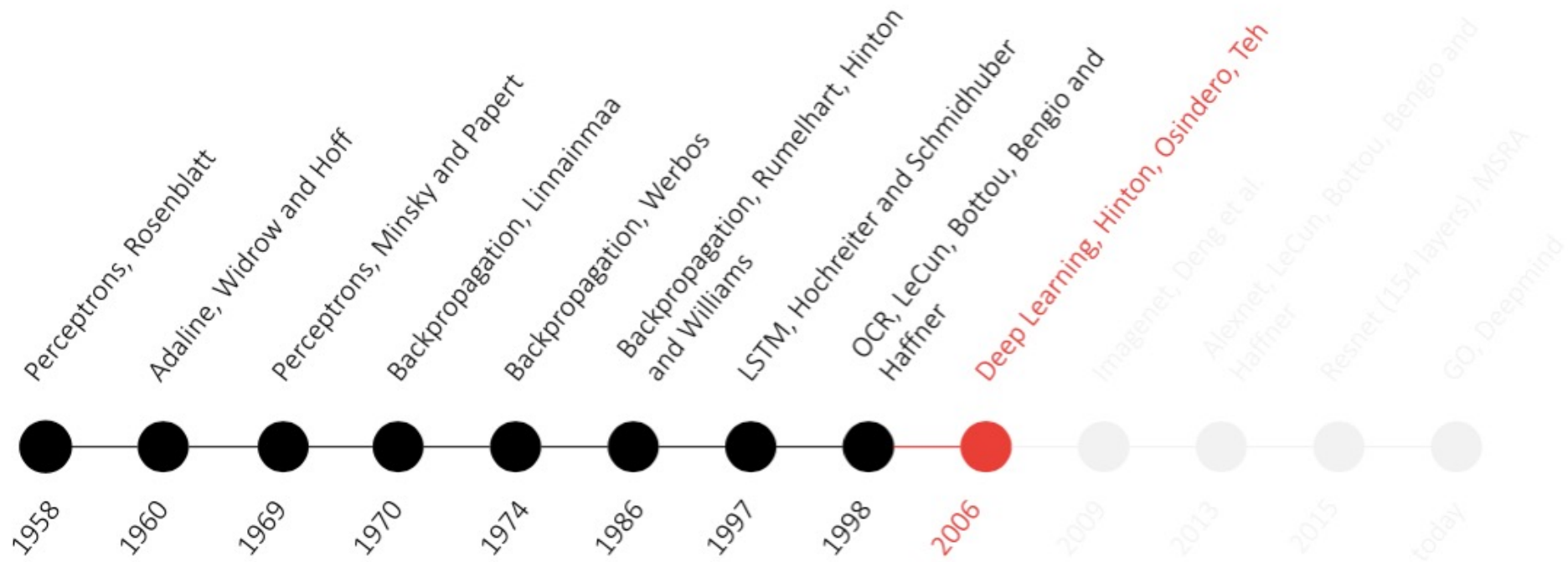
The second “AI winter”

- Until 1998 some nice algorithms and methods were proposed
 - Backpropagation
 - Recurrent Long-Short Term Memory Networks
 - OCR with Convolutional Neural Networks
- However, at the same time Kernel Machines (SVM etc.) suddenly become very popular
 - Similar accuracies in the same tasks
 - Neural networks could not improve beyond a few layers
 - Kernel Machines included much fewer heuristics & nice proofs on generalization
- As a result, once again the AI community turns away from Neural Networks

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The thaw of the “AI winter”



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Neural Network and Deep Learning problems

Lack of processing power

- No GPUs at the time

Lack of data

- No big, annotated datasets at the time

Overfitting

- Because of the above, models could not generalize all that well

Vanishing gradient

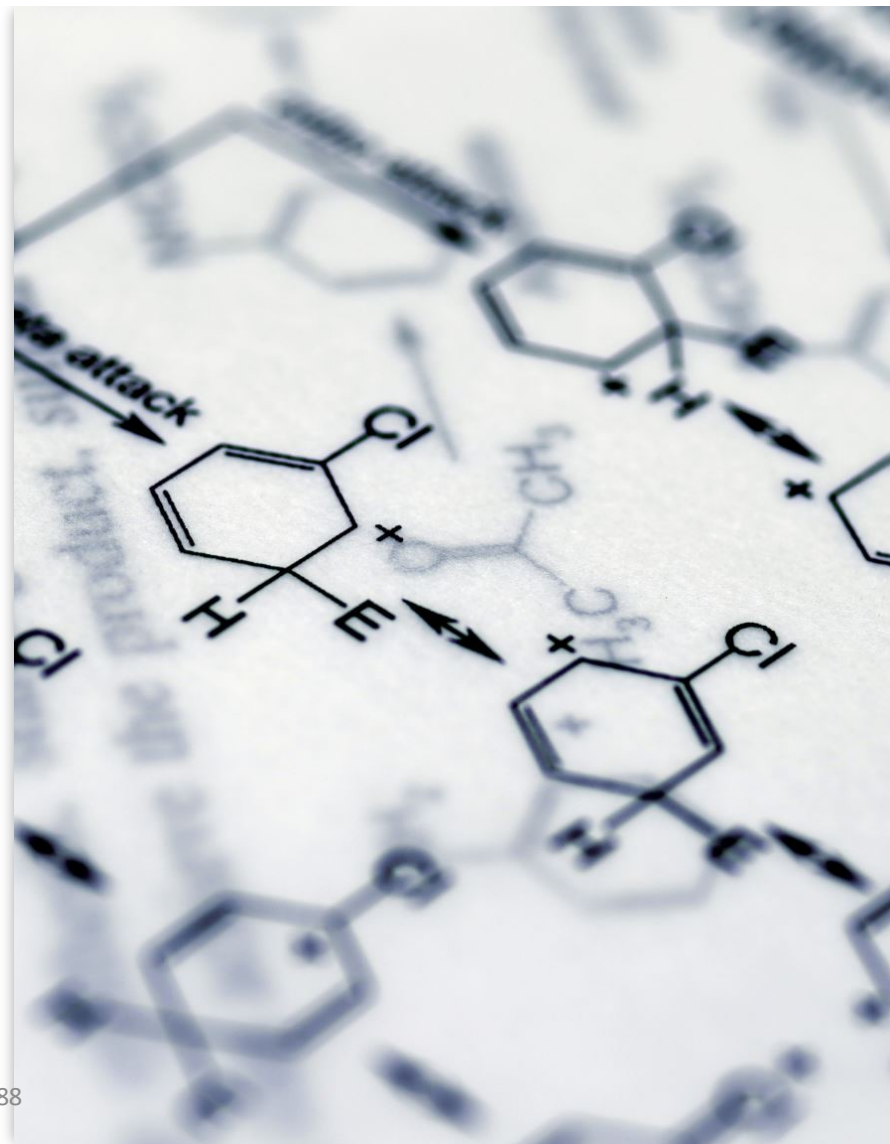
- While learning with NN, you need to multiply several numbers $a_1 \cdot a_2 \cdot \dots \cdot a_n$
- If all are equal to 0.1, for $n = 10$ the result is 0.0000000001, too small for any learning

Despite Backpropagation

...

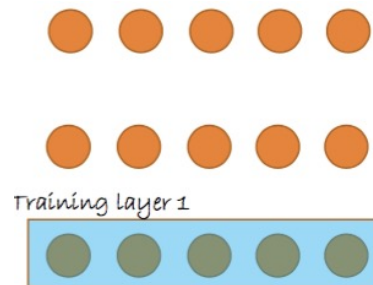
- Experimentally, training multi-layer perceptrons was not that useful
 - Accuracy didn't improve with more layers
- The inevitable question
 - Are 1-2 hidden layers the best neural networks can do?
 - Or is it that the learning algorithm is not really mature yet

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Deep Learning arrives

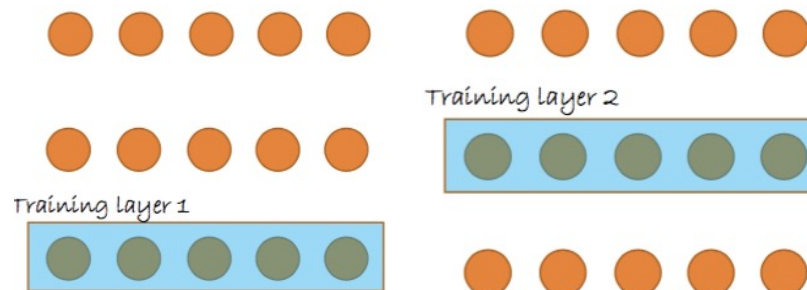
- Layer-by-layer training
 - The training of each layer individually is an easier undertaking
- Training multi-layered neural networks became easier
- Per-layer trained parameters initialize further training using contrastive divergence



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Deep Learning arrives

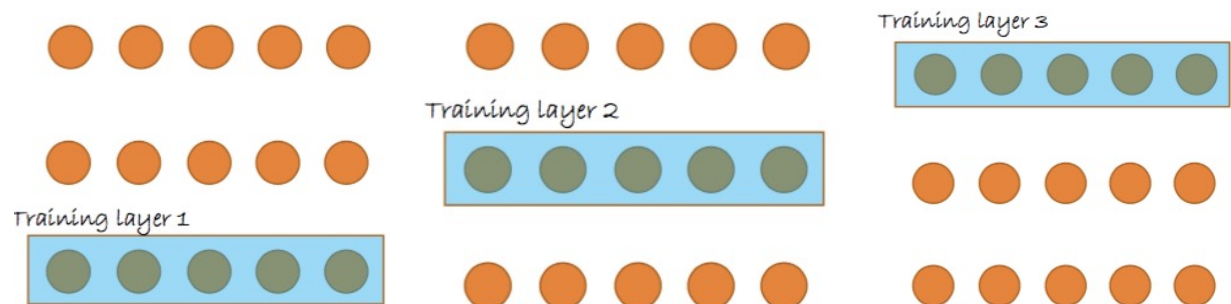
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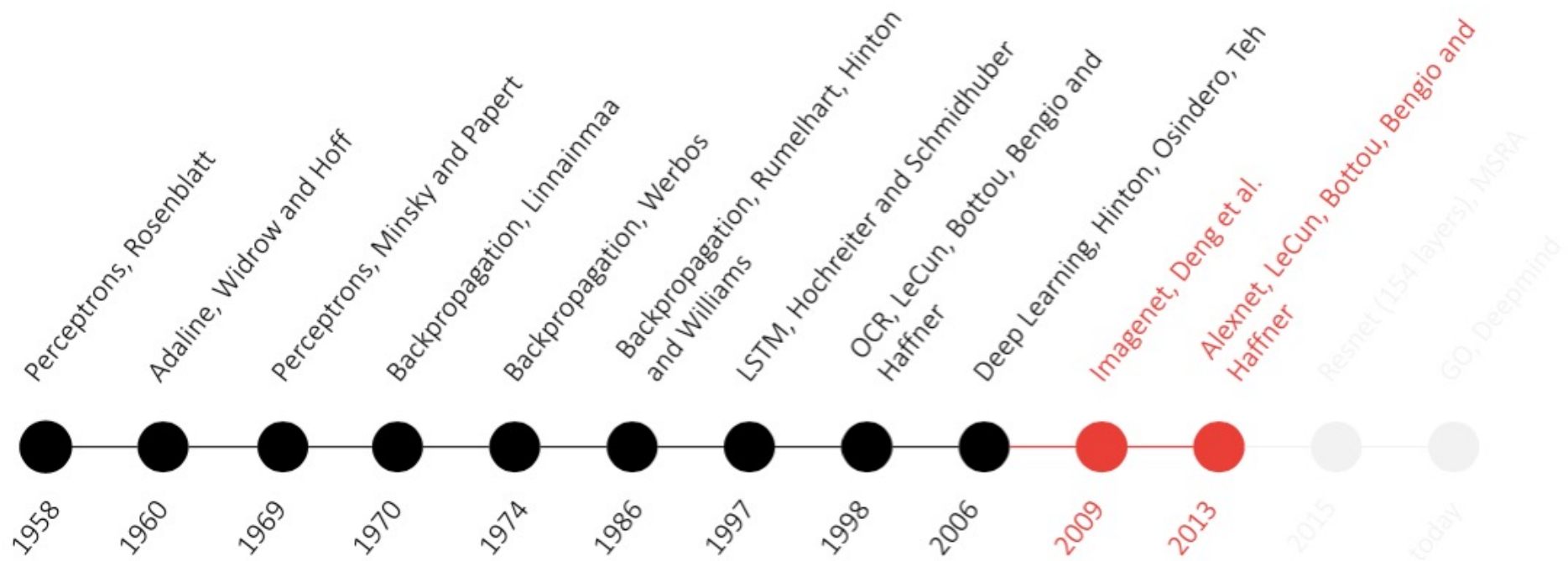
Deep Learning arrives

- Layer-by-layer training
 - The training of each layer individually is an easier undertaking
- Training multi-layered neural networks became easier
- Per-layer trained parameters initialize further training using contrastive divergence

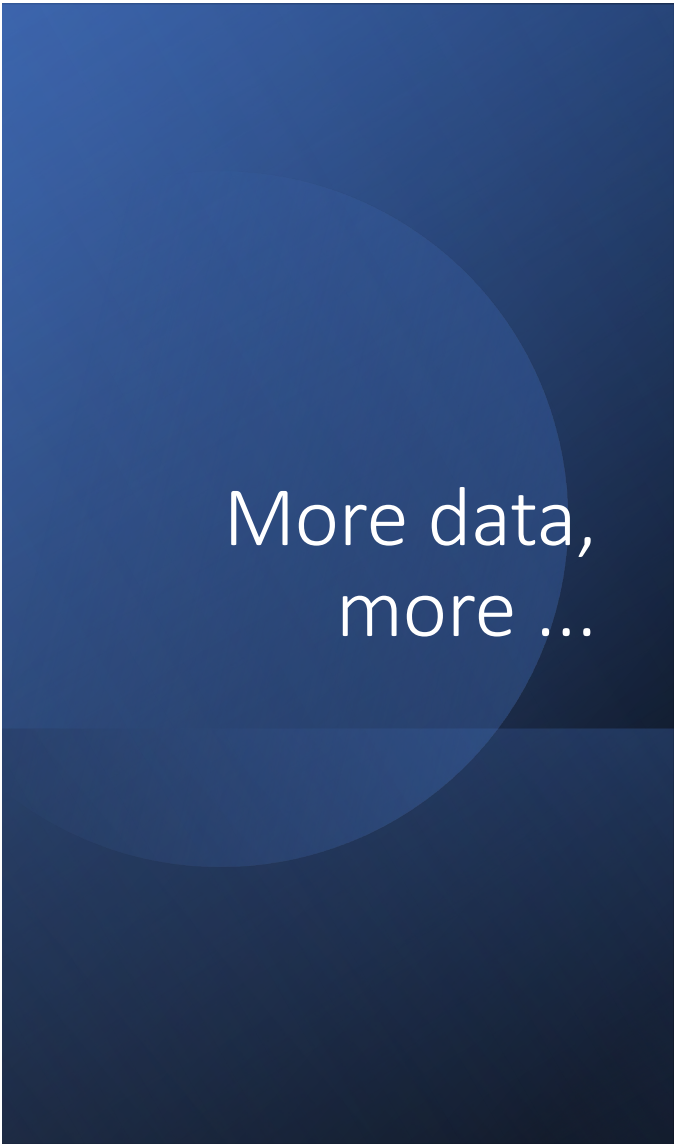


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Deep Learning Renaissance

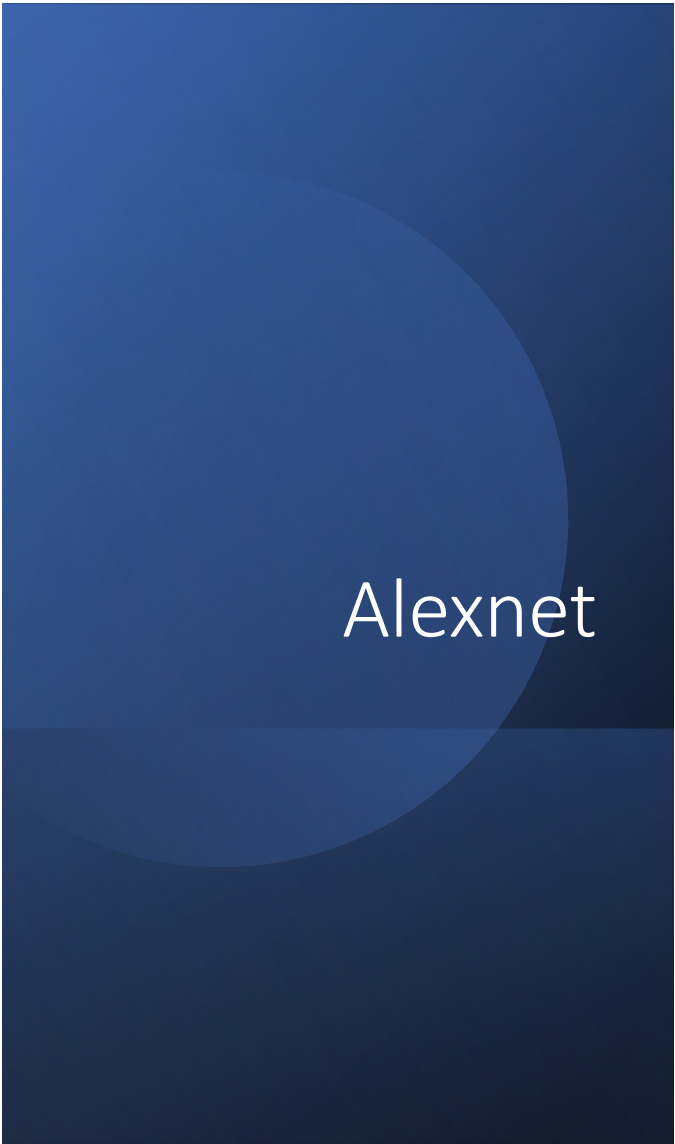


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More data,
more ...

- In 2009 the Imagenet dataset was published [Deng et al., 2009]
 - Collected images for each term of Wordnet (100,000 classes)
 - Tree of concepts organized hierarchically
 - “Ambulance”, “Dalmatian dog”, “Egyptian cat”, ...
 - About 16 million images annotated by humans
- Imagenet Large Scale Visual Recognition Challenge (ILSVRC)
 - 1 million images
 - 1,000 classes
 - Top-5 and top-1 error measured

The logo for Alexnet, featuring a dark blue background with a lighter blue circular shape on the left side. The word "Alexnet" is written in white text across the center of the circle.

Alexnet

- In 2013 Krizhevsky, Sutskever and Hinton re-implemented [Krizhevsky2013] a convolutional neural network [LeCun1998]
 - Trained on Imagenet, Two GPUs were used for the implementation
- Further theoretical improvements
 - Rectified Linear Units (ReLU) instead of sigmoid or tanh
 - Dropout
 - Data augmentation
- In the 2013 Imagenet Workshop a legendary turmoil
 - Blasted competitors by an impressive 16% top-5 error, Second best around 26%
 - Most didn't even think of NN as remotely competitive
- At the same time similar results in the speech recognition community
 - One of G. Hinton students collaboration with Microsoft Research, improving state-of-the-art by an impressive amount after years of incremental improvements [Hinton2012]

Alexnet architecture

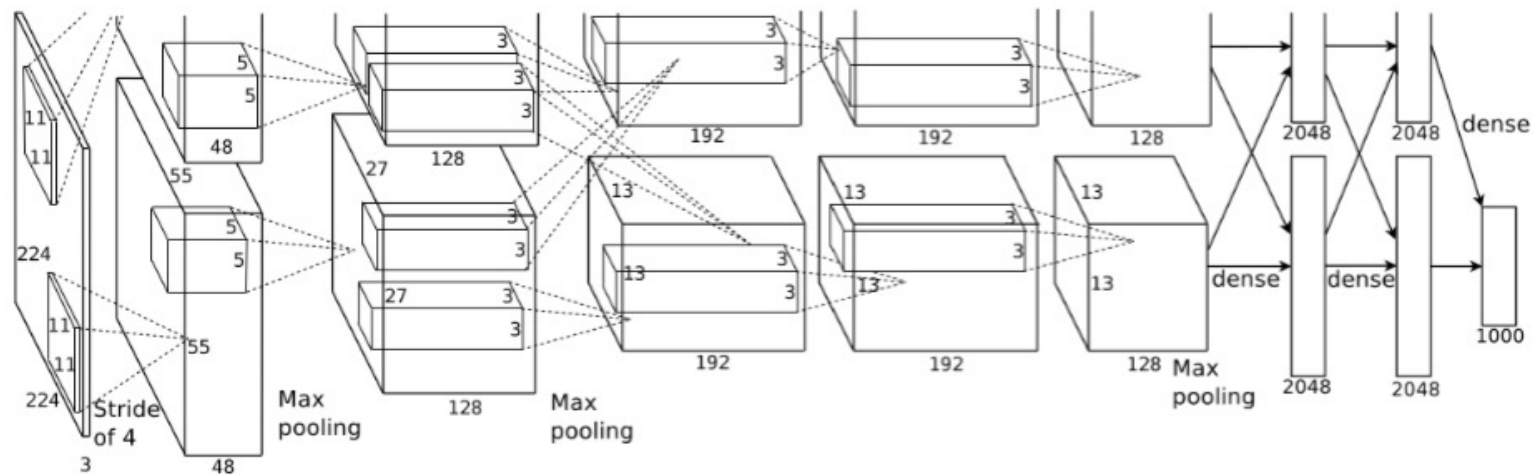
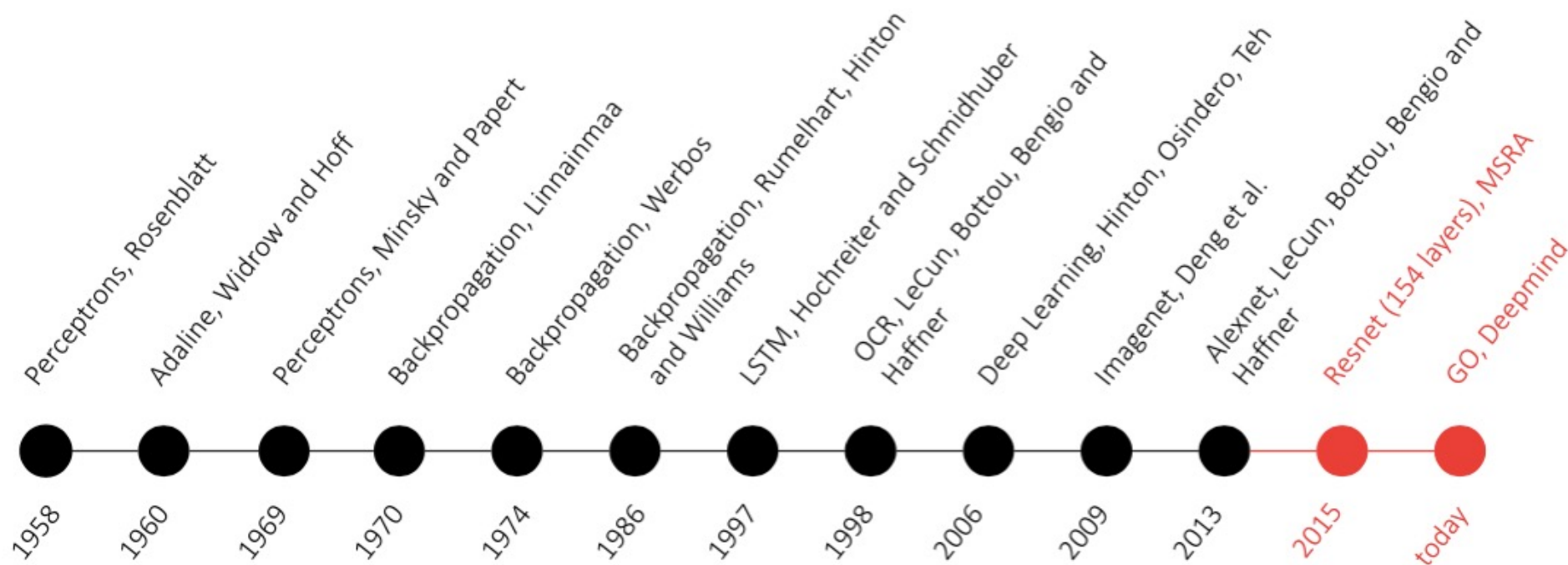


Figure 2: An illustration of the architecture of our CNN, explicitly showing the delineation of responsibilities between the two GPUs. One GPU runs the layer-parts at the top of the figure while the other runs the layer-parts at the bottom. The GPUs communicate only at certain layers. The network's input is 150,528-dimensional, and the number of neurons in the network's remaining layers is given by 253,440–186,624–64,896–64,896–43,264–4096–4096–1000.

Deep Learning Golden Era



The today

Deep Learning is almost everywhere

- Object classification
- Object detection, segmentation, pose estimation
- Image captioning, question answering
- Machine translation
- Speech recognition
- Robotics

Some strongholds

- Action classification, action detection
- Object retrieval
- Object tracking
- Chatbot

The ILSVC Challenge

2012 Teams	%error	2013 Teams	%error	2014 Teams	%error
Supervision (Toronto)	15.3	Clarifai (NYU spinoff)	11.7	GoogLeNet	6.6
ISI (Tokyo)	26.1	NUS (singapore)	12.9	VGG (Oxford)	7.3
VGG (Oxford)	26.9	Zeiler-Fergus (NYU)	13.5	MSRA	8.0
XRCE/INRIA	27.0	A. Howard	13.5	A. Howard	8.1
UvA (Amsterdam)	29.6	OverFeat (NYU)	14.1	DeeperVision	9.5
INRIA/LEAR	33.4	UvA (Amsterdam)	14.2	NUS-BST	9.7
		Adobe	15.2	TTIC-ECP	10.2
		VGG (Oxford)	15.2	XYZ	11.2
		VGG (Oxford)	23.0	UvA	12.1

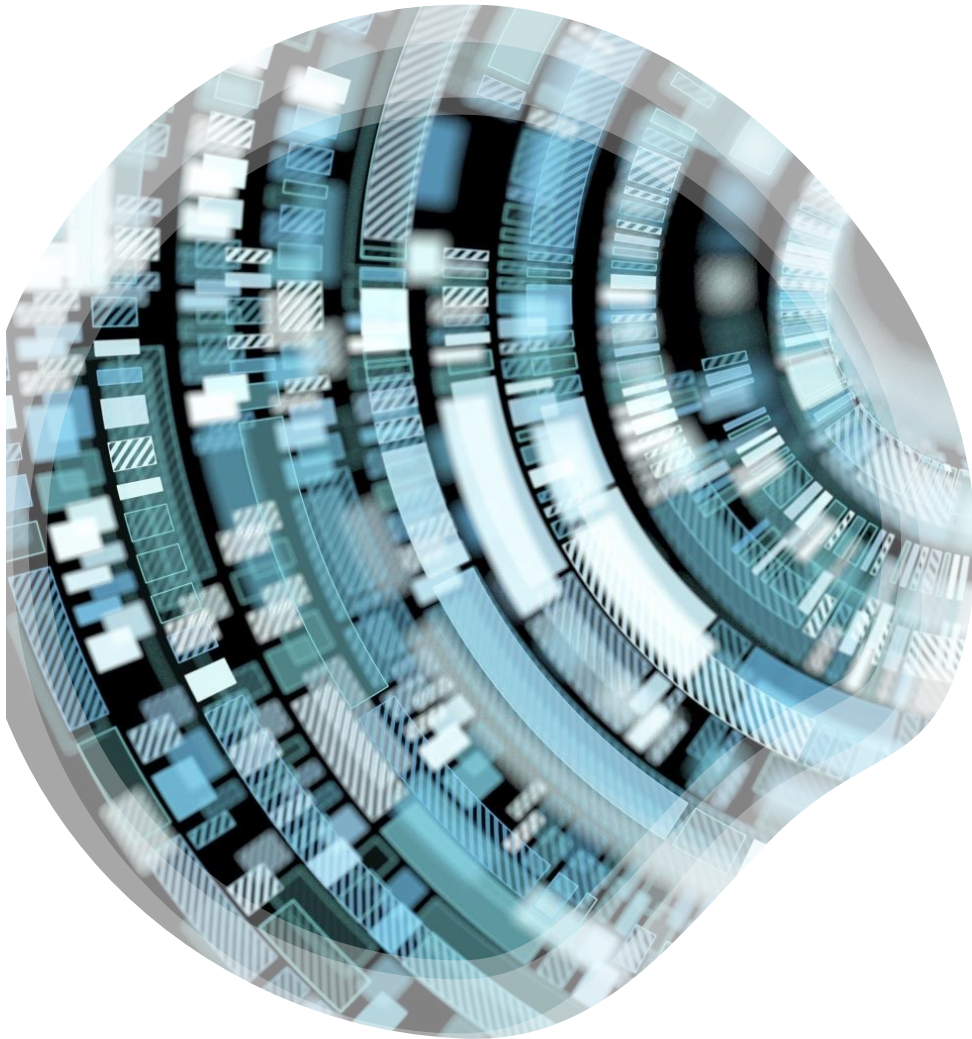
CNN based, non-CNN based

Figures taken from Y. LeCun's CVPR 2015 plenary talk



2015 ILSVRC Challenge

- Microsoft Research Asia won the competition with a legendary 150-layered network
- Almost superhuman accuracy: 3.5% error
 - In 2016 <3% error
- In comparison in 2014 GoogLeNet had 22 layers



Future?

- Artificial General Intelligence (AGI)
- Large language models
- Diffusion models (text-to-image)
- Neuro-Symbolic AI
- ...

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So, why now?



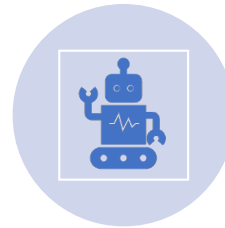
Better hardware



Bigger data

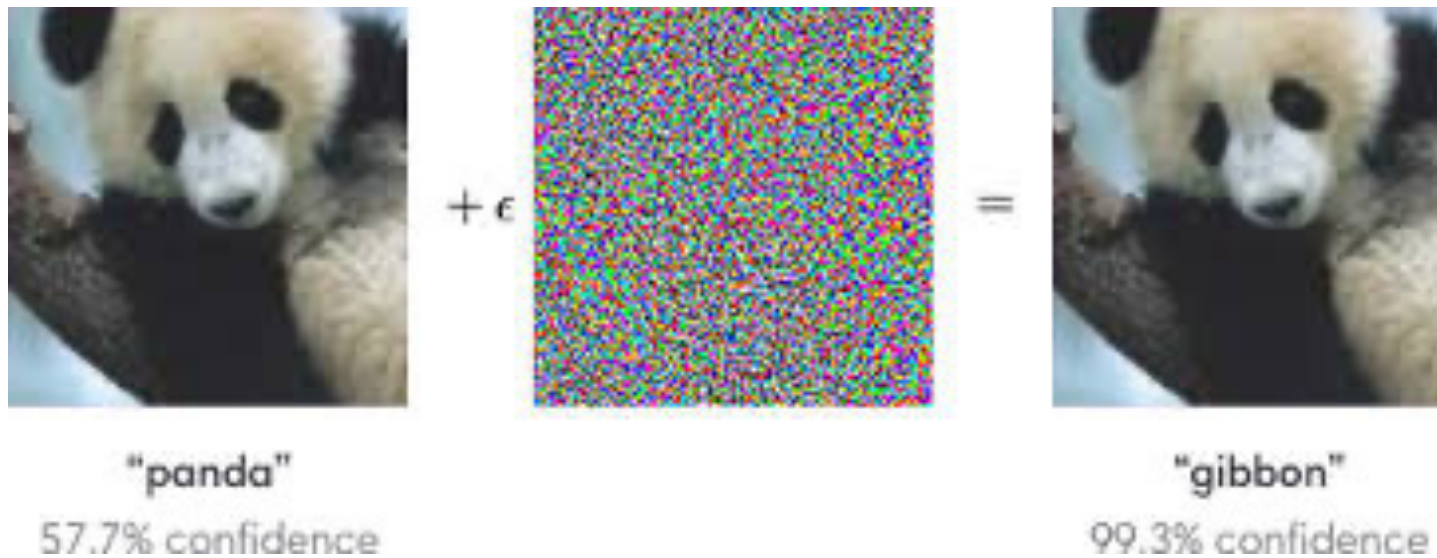


Better regularization
methods, such as
dropout



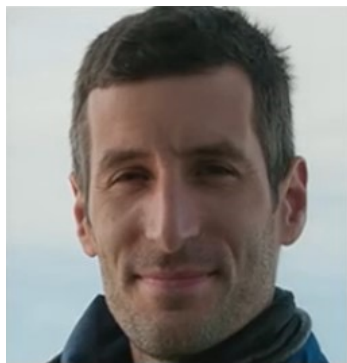
Better optimization
methods, such as Adam,
batch normalization

has deep learning no issue at all?



A small perturbation leads to class changes

has deep learning no issue at all?



+



2D printed eyeglasses

=



Anyone wearing this glasses will be
classified as

has deep learning no
issue at all?

- There are many more
- But we may be able to improve deep learning to counter these weaknesses – lots of research on this

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